Project Proposal Guidelines

MAT 219, Fall 2026

Your proposal is a short check-in, not a final commitment. You can still adjust your topic, dataset, or question after you get feedback. The goal is to make sure your group has a workable dataset and a question specific enough to actually answer with it, before you invest real time in it.

What to include

- **Group members**
  - Names of everyone in the group, and one point of contact.
- **Topic and dataset**
  - Which topic area you're using (from the project options list, or your own idea).
  - The specific dataset you plan to use, with a working link.
  - A sentence or two describing what's in it: roughly how many rows/records, what time period or region it covers, and how you accessed it (direct download, no login/API key required).
- **Variables**
  - List the specific variables (columns) from the dataset you plan to use.
  - For each one, note whether it's categorical or quantitative.
- **Research question(s)**
  - One or two specific, answerable questions (not just a topic area). "What affects X?" is a topic; "Is Y associated with X?" is a question.

> **Too vague:** "We'll study home prices and see what's interesting."
> **Better:** "Is the median county-level home price associated with median household income, and does that relationship look different for urban vs. rural counties?"

- **Feasibility check**
  - Confirm your group has actually opened/previewed the dataset and it loads the columns you expect. A dataset you've only seen described on a webpage doesn't count. Download it and look.

What you do NOT need yet

You have not learned the statistical methods you'll eventually use (confidence intervals, hypothesis tests, regression, etc.). At the proposal stage, just describe in plain language what kind of comparison you expect to make: e.g., "compare two groups," "look for a relationship between two numerical variables," or "describe how a variable is distributed across regions." You'll choose the specific method later in the course, and it's okay if that choice changes.

Format

- About 1 page (not including any dataset links or references), plus your variable list (which can be a short bulleted list, not prose).
- Submit as a single file per group on Canvas.
- Due October 5th. Late proposals lose points under the standard late policy.

How the proposal is graded

This will be graded on whether…	✓
The research question is specific enough to be answered with the proposed dataset (not just a general topic)	☐
The dataset is real, accessible, and matches what's described	☐
Variables are correctly identified and correctly classified as categorical or quantitative	☐
The group has verified the dataset actually opens and contains the expected variables	☐
Proposal is within the length guideline and submitted on time	☐

A note on using AI tools

You may use AI tools to help you find datasets or brainstorm questions, but the proposal should reflect your own group's understanding of the data you chose. If you can't explain in your own words what your variables mean or why your question is answerable with them, the proposal isn't ready to submit.